



Computer Networks 01 | Introduction to IP Addressing | Computer Science and IT | GATE Crash Course

Generated on October 23, 2024

Summary

AI Notes

AI Slides

Text Notes

Screenshots

✦ 0

✦ 0

☐ 0

📷 13

Basics of IP Addressing

$2^1 = 2$ ✓	$2^{40} = 1 \text{ T (Tera)}$
$2^2 = 4$ ✓	$2^{50} = 1 \text{ P (Peta)}$
$2^3 = 8$ ✓	$2^{60} = 1 \text{ E (Exa)}$
$2^4 = 16$ ✓	$2^{70} = 1 \text{ Z (Zetta)}$
⋮	$2^{80} = 1 \text{ Y (Yotta)}$
$2^9 = 512$ ✓	
$2^{10} = 1024 = \underline{1\text{K}}$ (Kilo)	
$2^{20} = 1024 \times 1024 = \underline{1\text{M}}$ (Mega)	
$2^{30} = 1024 \times 1024 \times 1024 = \underline{1\text{G}}$ (Giga)	

▶ 20:50

Basics of IP Addressing

1 Byte = 8 bits

1 KB = 1024 Byte

1 MB = 1024 KB (Kilo Byte)

1 GB = 1024 MB (Mega Byte)

1 TB = 1024 GB (Giga Byte)

1 PB = 1024 TB (Tera Byte)

1 EB = 1024 PB (Peta Byte)

1 ZB = 1024 EB (Exa Byte)

1 YB = 1024 ZB (Zetta Byte)

Bits → b
Byte → B

▶ 23:16

Basics of IP Addressing

Binary Number	Decimal Value
00000000 → 0	
00000001 → 1	
00000011 → 3 $[2^2 - 1]$	
00000111 → 7 $[2^3 - 1]$	
00001111 → 15 $[2^4 - 1]$	
00011111 → 31 $[2^5 - 1]$	
00111111 → 63 $[2^6 - 1]$	
01111111 → 127 $[2^7 - 1]$	
11111111 → 255 $[2^8 - 1]$	

8 bit
Range → 0 to $2^8 - 1$

n bit
Range → 0 to $2^n - 1$

▶ 30:41

Basics of IP Addressing

Binary Number

Decimal Value

128 64 32 16 8 4 2 1
 10000000 → 128
 11000000 → 192
 11100000 → 224
 11110000 → 240 [255-15]
 11111000 → 248 [255-7]
 11111100 → 252 [255-3]
 11111110 → 254 [255-1]
 11111111 → 255

128
 64
 192
 32
 224
 16
 240

▶ 37:19

2 bit

00

01

10

11

3 bit

000

001

010

011

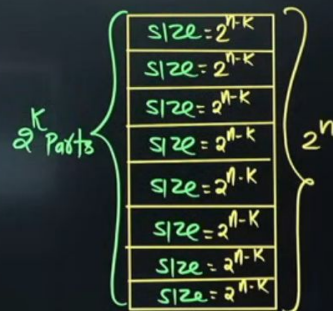
100

101

110

111

..... n bit

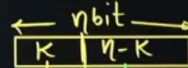


1 bit → 2 Parts = 2^1 Parts

2 bits → 4 Parts = 2^2 Parts

k bit → 2^k Parts

2^k Parts → 2^n
 1 Parts → 2^n
 $= \frac{2^n}{2^k} = 2^{n-k}$



2^k Parts
 2^{n-k} size of each Parts

▶ 49:19

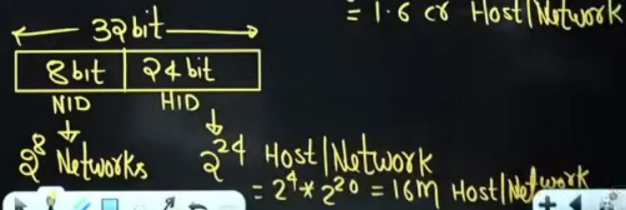
Introduction to IP Addressing

IPv4 Address = 32 bit $\rightarrow 2^8 * 2^{30} = 4G$ IP Addresses

Total number of IP addresses = $2^{32} = 4,294,967,296$

Initially in 1980's IP Addresses was divided into two Fixed Parts i.e.,

NID = 8 bit, and HID = 24 bit.



▶ 58:30

Introduction to IP Addressing

32 bit

8	24
---	----

2^8 Parts 2^{24} (size of each Part)
 256 Parts

$N_1 = 2^{24}$ IP
$N_2 = 2^{24}$ IP
$N_3 = 2^{24}$ IP
N_4 "
N_5 "
"
"
"
N_{256} "

2^8 Parts
 256 Parts
 256 Networks

2^{32} IP Addresses

ENALLAH

▶ 1:15:01

Introduction to IP Addressing

Disadvantage

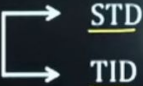
There are only 256 Network's, and even a small organization must buy 16M computer (HOST) to purchase one network.

Soln: classful Addressing

GATE WALLAH

▶ 1:18:05

Telephone Networks

1. 11 digit Number ✓
2. Two Parts 
 - STD
 - TID
3. Unique ✓

GATE WALLAH

▶ 1:19:50

Telephone Networks

City	
3	8
<u>STD</u>	<u>TID</u>

011 2416191627

Town	
4	7
<u>STD</u>	<u>TID</u>

0120 2495271

Villages	
5	6
<u>STD</u>	<u>TID</u>

05924 264297

▶ 1:22:03

Computer Networks

1. 32 bit Number $\left[\boxed{8} \cdot \boxed{8} \cdot \boxed{8} \cdot \boxed{8} \right]$
2. Two Parts $\begin{cases} \rightarrow \text{NID} \\ \rightarrow \text{HID} \end{cases}$
3. Unique ✓

▶ 1:30:46

Computer Networks

Class A	
8 bit	24 bit

NID $\downarrow$ $2^8 = 256$ Networks
 HID $\downarrow$ 2^{24} Host/Network

Note: Designed $= 2^4 \times 2^{20} = 16M$ Host/Network
 $\approx 1.6 \text{ cr}$ Host/Net.
 For Big organization
 For example NASA

Class B	
16 bit	16 bit

NID $\downarrow$ 2^{16} N/w's = 64K Host/Net.
 HID $\downarrow$ 2^{16} Host/Net.
 $= 2^6 \times 2^{10}$
 $= 64K$ Networks

Note: Designed For MNC's
 For ex: IBM, Wipro, TCS etc.

Class C	
24 bit	8 bit

NID $\downarrow$ 2^{24} N/w
 HID $\downarrow$ 2^8
 $= 256$ Host/Net.
 $2^4 \times 2^{20}$
 $= 16M$ Networks
 $\approx 1.6 \text{ cr}$ Network

▶ 1:47:14

Class C

Designed For small organizations For example schools, colleges etc

▶ 1:47:15